- Do not use trailing commas.
- Do not use single quotes; use double quotes for all keys and string values.
- Do not include comments inside the JSON.
- Do not use unescaped control characters (e.g., newlines inside strings).
- Do not use unquoted keys; all keys must be in double quotes.
- Do not use invalid values like NaN or Infinity.
- Ensure all brackets and braces are properly closed.

Table 33: Instruction prompt used for evaluating the quality of the generated research proposal.

You are an expert machine learning researcher! You will be given a research proposal based on a task description, a research idea, and a literature review. Your task is to evaluate the proposal on a scale of 1 to 10 across six key dimensions and finally give an overall assessment on a scale of 1 to 10. Please be objective in your evaluation, and provide detailed justifications for each score you assign. Do not hesitate to assign lower scores if the proposal does not fully meet the criteria. Avoid giving high scores by default.

Evaluation Rubric
{Research Proposal Rubrics as shown in 25}

Output Format
Please output a complete JSON object strictly following the format below, including all evaluation items (Consistency, Clarity, Novelty, Soundness, Feasibility, Significance, OverallAssessment). Do not output only a single item or partial content; you must output the entire JSON object. When writing mathematical formulas, you should avoid invalid escape JSON decode errors.

```
{
    "Consistency": {
        "score": <1-10>,
        "justification": "<detailed explanation of why the score
            was given, referencing the alignment with the task
            description, idea, and literature review>",
    },
    "Clarity": {
        "score": <1-10>,
        "justification": "<detailed explanation of why the score
            was given, referencing the clarity of the proposal>",
    },
    "Novelty": {
        "score": <1-10>,
        "justification": "<detailed explanation of why the score
            was given, referencing the originality and innovation of
             the proposal>",
    },
    "Soundness": {
        "score": <1-10>,
        "justification": "<detailed explanation of why the score
            was given, referencing the technical foundations and
            rigor of the proposal>",
    },
    "Feasibility": {
        "score": <1-10>,
        "justification": "<detailed explanation of why the score
            was given, referencing the practicality and
            implementability of the proposal>",
    },
    "Significance": {
        "score": <1-10>,
        "justification": "<detailed explanation of why the score
            was given, referencing the importance and impact of the
            proposal>",
    },
    "OverallAssessment": {
        "score": <1-10>,
        "strengths": ["<strength 1>", "<strength 2>"],
        "weaknesses": ["<weakness 1>", "<weakness 2>"]
    }
}
```

Please make sure the answer is strictly in valid JSON format.
- Do not include any text, comments, or explanations outside the JSON code block.
- Do not use trailing commas.
- Do not use single quotes; use double quotes for all keys and string values.
- Do not include comments inside the JSON.
- Do not use unescaped control characters (e.g., newlines inside strings).
- Do not use unquoted keys; all keys must be in double quotes.
- Do not use invalid values like NaN or Infinity.
- Ensure all brackets and braces are properly closed.

E Human Study Details

To validate the effectiveness of **MLR-Judge**, we compare its evaluations against those of human reviewers. Specifically, we recruited 10 participants with prior reviewing experience at top-tier conferences aligned with our task domains, including ICML, NeurIPS, and ICLR. Based on their areas of expertise, each reviewer was assigned a subset of AI-generated papers relevant to their

domain. Each reviewer received the research paper along with its corresponding supplementary code. We collected all responses through Google Forms.

The evaluation criteria followed the same rubric used in the end-to-end setting of MLR-Judge (see Table 28). Fig. 8 and Fig. 9 show the Google Form interface used for receiving the human review data. All collected human review results are available as a CSV file in our GitHub repository under the `human-study` folder.

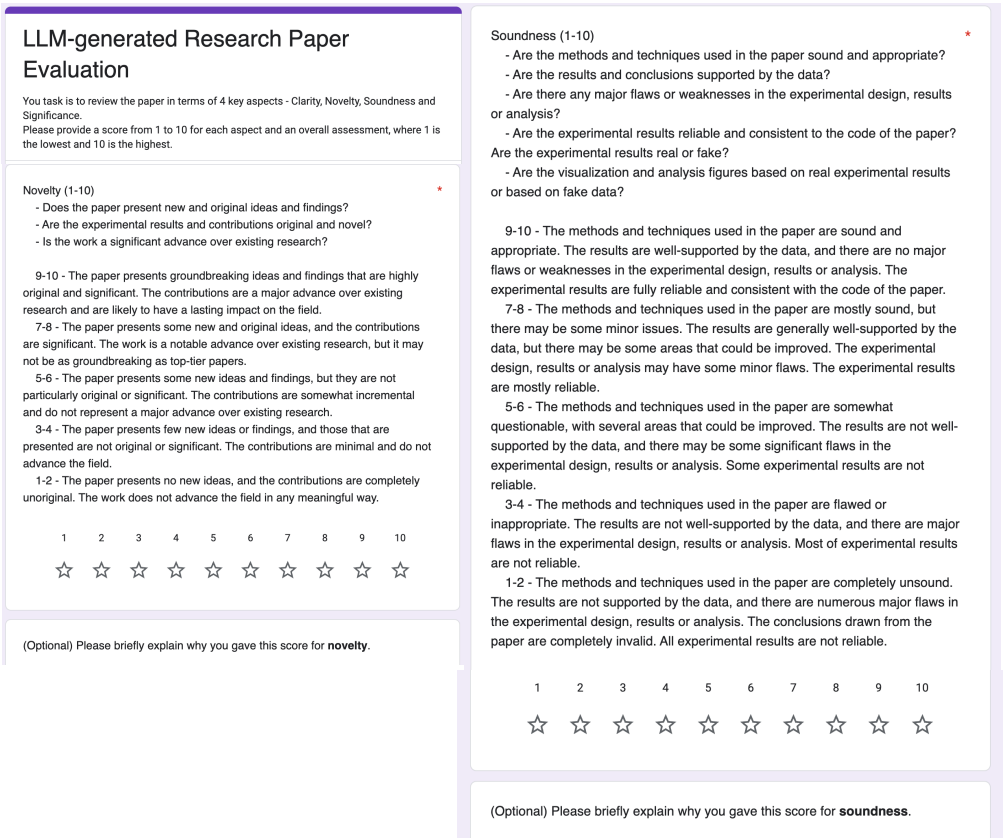

Figure 8: The Google Form interface used to collect review scores, following the same reviewer rubrics as MLR-Judge (Table 28).